

Lecture Summariser — Proposal

Canonical proposal. A mirror lives in `Project Hub/projects/proposal_Lecture Summariser.md` for planning. Converted from `LLM lecture note taker.docx` on 2026-06-02.

Lecture Summariser — Project Roadmap

Phase 1 — MVP Foundation

The goal here is a working end-to-end pipeline with no bells and whistles.

Inputs: PDF slide deck + plain text transcript

Deliverables:

1. Slide parser — converts PDF pages to images and extracts raw text per slide
2. Visual region detector — identifies large non-text regions (diagrams, figures, charts) worth keeping
3. Transcript chunker — splits transcript into segments that can be assigned to slides
4. Basic aligner — maps transcript chunks to slides using keyword overlap and slide order
5. Note generator — prompts an LLM to produce structured notes per slide section using the aligned text
6. LaTeX exporter — outputs a compilable .tex file with sections, text, and embedded slide images

Success criterion: Given a lecture, produce a readable, structured .tex document that a student could use for revision.

Phase 2 — Multimodal Grounding

The goal here is making the image-context link meaningful, which is the core novelty of the project.

1. Improved visual region detection — filter out decorative elements, logos, and backgrounds; retain only pedagogically useful visuals
2. Per-figure context extraction — for each extracted figure, retrieve the transcript span that explains it and generate a short caption or description
3. Semantic alignment — replace keyword-based alignment with embedding similarity so the system handles cases where the lecturer paraphrases or elaborates beyond the slide text
4. Text-image grounding in notes — figures appear inline next to the explanation that corresponds to them, not just appended at the end of a section

Success criterion: Each extracted figure in the output has a contextually accurate explanation drawn from the transcript, not just the slide bullet text.

Phase 3 — Richer Output and Quality

1. Equation extraction and formatting — detect mathematical content on slides and format it properly in LaTeX
2. OCR for diagram labels — read axis labels, annotations, and callouts within figures to improve captions
3. Note quality refinement — add key takeaways, definitions, and example callouts where the transcript supports them
4. Revision mode vs full notes mode — shorter bullet-driven output vs longer prose explanation, selectable at export time

Phase 4 — Extension (if time allows)

1. Flashcard generation — derive question-answer pairs from grounded notes for spaced repetition
2. Multi-lecture support — handle a series of lectures and produce a unified document with consistent section numbering
3. Slide format flexibility — support .pptx input directly, not just PDF

Core Technical Challenges (in priority order)

1. Deciding which visual regions are worth extracting
2. Aligning transcript spans to the correct slide and figure
3. Generating captions that reflect what the lecturer said, not just what the slide shows
4. Keeping the LaTeX output clean and compilable across varied inputs

Module Summary

Module	Phase
Slide parser	1
Visual region detector	1 → refined in 2
Transcript chunker	1
Transcript-slide aligner	1 → semantic in 2
Per-figure context extractor	2
Note generator	1 → richer in 3
LaTeX exporter	1
Equation / OCR extraction	3
Flashcard generator	4